

Growth Theory

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Introduction

So far we've been analyzing a **static picture** of the economy.

Our closed economy model, for example, analyzes the economy in a specific point in time.

The same is true for the monetary system we've been analyzing.

We are shifting gears now. We will analyze the economy in a **dynamic setting**.

We will go from analyzing a picture to analyzing a movie.

Introduction

Average income (real GDP per person) varies hugely between different countries.

Country	Real GDP per capita 2022 (PPP)
United States	58,487
Germany	46,648
South Korea	41,321
United Kingdom	38,407
Japan	38,269
Mexico	16,235
Brazil	14,640
India	7,766
Ethiopia	2,289
Mozambique	1,088

Source: Bolt and van Zanden - Maddison Project Database 2023

Average income in the US is **more than 20 times higher** than in Ethiopia!

These large differences in income are reflected in **large differences** in “**quality of life**”.

Introduction

Growth rates of GDP per capita also vary substantially among countries.

Average Growth Rate of GDP per Capita: 1990 - 2022

Country	Growth Rate	Years to Double GDP per Capita
United States	1.44%	48.6
Germany	1.91%	37.2
South Korea	3.47%	20.9
United Kingdom	1.20%	60.0
Japan	0.77%	90.9
Mexico	1.62%	43.2
Brazil	1.96%	36.7
India	4.19%	17.2
Ethiopia	2.91%	24.0
Mozambique	-1.52%	-

Source: Bolt and van Zanden - Maddison Project Database 2023

Introduction

The last table shows the impact of **small differences** in **growth rates** over time.

Take US and Germany as an example: the difference in their growth rates is just 0.47%.

- Germany doubles its GDP per capita every 37.2 years.
- The US doubles its GDP per capita every 48.6 years.
- At this rate, Germany will surpass the US in GDP per capita in less than 50 years.

Inspired by these facts, in this lecture we will try to answer the following questions:

- What **explains these differences** in income and growth rates?
- How can the **rich countries** be sure to **maintain** their high **standard of living**?
- What **policies** should the **lower income countries** pursue to **promote** more **rapid growth**?

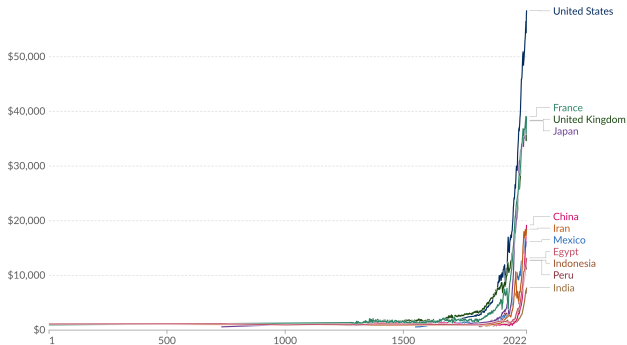
Before that, let's see a bit of the **history of economic growth**.

Introduction

GDP per capita, 1 to 2022

This data is adjusted for inflation and for differences in the cost of living between countries.

Our World
In Data



Data source: Bolt and van Zanden - Maddison Project Database 2023

OurWorldInData.org/economic-growth | CC BY

Note: This data is expressed in international-\$¹ at 2011 prices.

1. International dollars: International dollars are a hypothetical currency that is used to make meaningful comparisons of monetary indicators of living standards. Figures expressed in international dollars are adjusted for inflation within countries over time, and for differences in the cost of living between countries. The goal of such adjustments is to provide a unit whose purchasing power is held fixed over time and across countries, such that one international dollar can buy the same quantity and quality of goods and services no matter where or when it is spent. Read more in our article: What are Purchasing Power Parity adjustments and why do we need them?

Introduction

Last picture shows the evolution of GDP per capita over time.

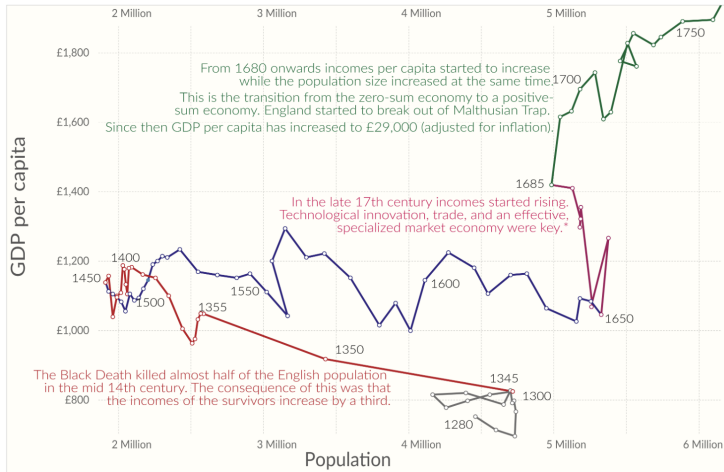
For most of the history, the standards of living were extremely low, not much different from that of Ethiopia today.

Until 1700 (pre-growth economy), the world was in a **Malthusian trap**.

- There was a tight **link** between **population growth** and **living standards**.
- More births, lower incomes.
- More deaths, higher incomes.
- A **zero-sum game**:
 - The size of the pie (total output) was not changing much.
 - The only way to get a bigger slice was to take it from someone else.

The best way to understand the Malthusian trap is to see the effects of the **Black Death** in England in the 14th century.

Introduction



Data: Broadberry, S., Campbell, B. M., Klein, A., Overton, M., & Van Leeuwen, B. (2015). British Economic Growth, 1270–1870 and Bank of England for the period thereafter. Data prior to 1700 refers to England; data thereafter refers to the UK. Averages over 5 year intervals are shown here.

*An additional factor was another pandemic, the Great Plague of London hit England in 1665 and reduced population size.

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Introduction

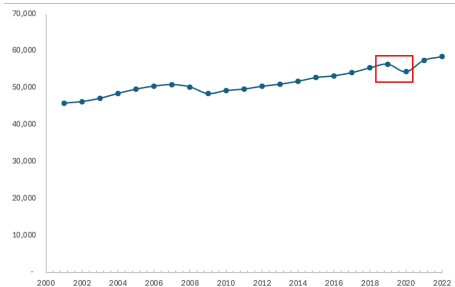
The Black Death killed about 50% of the population of England.

However, GDP per capita increased by $\frac{1}{3}$ in the period.

This is an extremely different world from the one we live in today.

Remember the COVID-19 pandemic? Look what happened to GDP per capita in the US:

US Real GDP per capita



Introduction

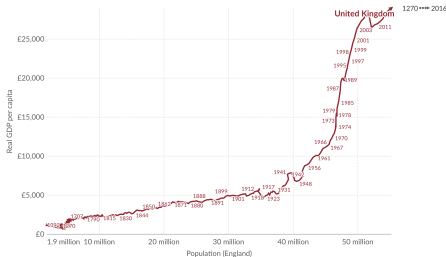
The first country to break free from the Malthusian trap was **England**.

Starting in 1685, England experienced a **sustained increase** in **living standards**.

For the first time, **the speed of innovation** became so fast that the size of the **population** and **income per capita** started to **grow** at the **same time**.

England left the zero-sum game economy and entered a **positive-sum game** economy.

England's economy over the long run: GDP per capita vs. population, United Kingdom, 1270 to 2016



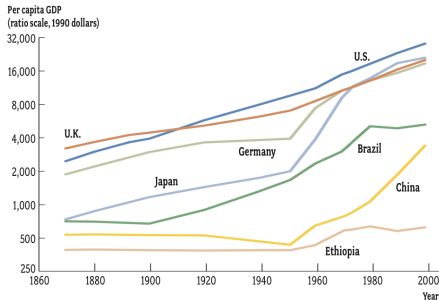
Data source: A Millennium of Macroeconomic Data - Bank of England

OurWorldInData.org/breaking-the-malthusian-trap | CC BY

Introduction

After England, sustained growth emerged in many other countries, at different times.

Per Capita GDP in Seven Countries, 1870–2000



Source: Maddison, *The World Economy* (see Figure 3.1). Observations are presented every decade after 1950 and less frequently before that as a way of smoothing the series.

Arguably, the **Industrial Revolution** was the **most important event** in the history of humanity since the domestication of animals and plants.

Introduction

Unfortunately, a large part of the world is still living on very low incomes.

Understanding the sources of economic growth is crucial to help these countries.

We will start our journey by analyzing a model of economic growth called the **Solow model**.

The Solow model

Our task in this section is to develop a theory of economic growth called **the Solow Growth Model**.

The model is named after Robert Solow, who won the Nobel Prize in Economics in 1987 for his work on this model.

The model shows how savings, population growth and technological progress affect the economy's output and growth rate.



Robert M. Solow in 1987, when he won the Nobel Memorial Prize in Economic Sciences. Mark Lennihan/Associated Press

The Solow model

We will start by analyzing the role of saving!

For now, there's no population growth and no technological progress.

Capital accumulation will be the engine of growth in this model!

Let's see the assumptions of the model.

Solow Model: Supply of Goods

The economy produces a **single good**.

The output is described by a production function F :

$$Y = F(K, L)$$

where Y is output, K is capital and L is labor.

F has constant returns to scale (**CRS**): $F(zK, zL) = zF(K, L)$

CRS allows us to **analyze** all **quantities** in the economy relative to the size of the **labor force**. In the definition of CRS, let $z = 1/L$:

$$F\left(\frac{1}{L} \cdot K, \frac{1}{L} \cdot L\right) = \frac{1}{L} \cdot F(K, L) = \frac{1}{L} \cdot Y$$

This means that we can analyze the economy in terms of per worker (or per capita) variables.

Solow Model: Supply of Goods

We will refer to **per capita** (or per worker) quantities with **lowercase letters**.

Then, output per worker is:

$$y = Y/L = F(k, 1) = f(k)$$

where:

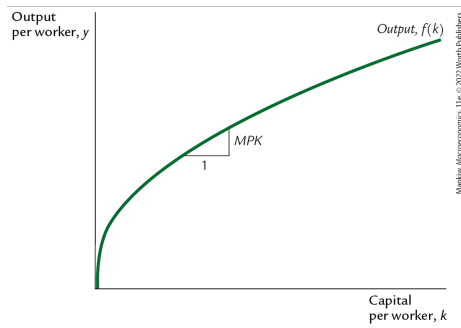
- $f(k) = F(k, 1)$ is the per worker production function
- y is output per worker
- k is capital per worker

CRS implies that **L does not affect** the **relationship** between **y** and **k** !

Example: Suppose the production function is $F(K, L) = K^\alpha L^{1-\alpha}$. What is the per worker production function?

Solow Model: Supply of Goods

The beautiful production function $f(k)$:



Remember your old friend **MPK**:

- It is the **slope** of the production function.
- It tells us how much output increases when we increase capital by a little bit.
- We will assume diminishing marginal product of capital, i.e, **MPK is decreasing!**

Solow Model: Demand for Goods

Demand for goods comes from consumption and investment:

$$y = c + i$$

Output per worker is divided between:

- c consumption per worker.
- i investment per worker.

This equation is the per worker version of the national income identity:

- It omits government spending, which we ignore for simplicity.
- It omits net exports, since we assume a closed economy.

We will assume that people save a constant fraction s of their income.

Solow Model: Demand for Goods

Then, they must consume a fraction $1 - s$ of their income:

$$c = (1 - s) \cdot y$$

We call s the **savings rate**.

- It is a fraction, so it varies between 0 and 1!
- Many things can influence s such as interest rates, income, etc.
- For now, we will assume that **s is constant** and is given (exogenous)!

How much is investment per worker then?

$$i = y - c = y - (1 - s)y = s \cdot y$$

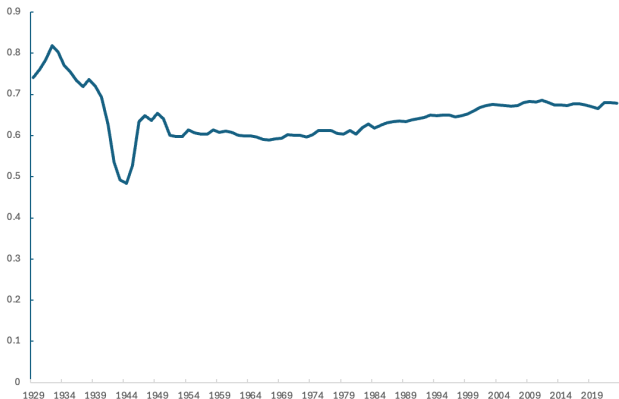
Nothing new here! **Savings equals investment**, always!

We can also refer to s as the **investment rate**!

Savings

Assuming people save a constant fraction of their income is a **simplification**.
How good is this assumption?

Consumption as a fraction of income in the US



At least in the **aggregate**, the assumption seems to hold!

Capital accumulation

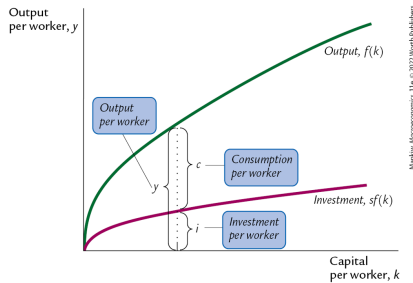
The **capital stock determines** the level of **output** in the Solow model.

Then, **changes in capital** stock can lead to **economic growth**.

Two forces determine the change in the capital stock: **investment** and **depreciation**.

Investment: $i = s \cdot y = s \cdot f(k)$

- Expenditure on new plant and equipment.
- Increases the capital stock.

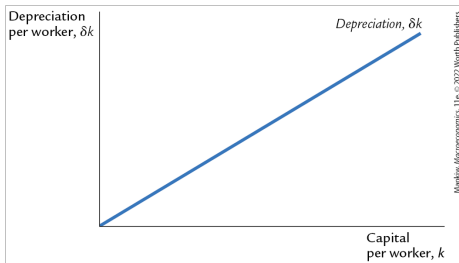


Capital accumulation

Depreciation:

- The wearing out of old capital.
- Decreases the capital stock.
- We assume that a fraction δ of the capital stock wears out each year.
- δ is the **depreciation rate**.

$$\text{Depreciation} = \delta k$$



Capital accumulation

The change in the capital stock is:

Change in Capital Stock = Investment – Depreciation

$$\Delta k = i - \delta k$$

$$\Delta k = sf(k) - \delta k$$

Now suppose you start with a capital stock k_0 at time $t = 0$.

The capital stock at time $t = 1$ is:

$$k_1 = k_0 + \Delta k_0 = k_0 + sf(k_0) - \delta k_0$$

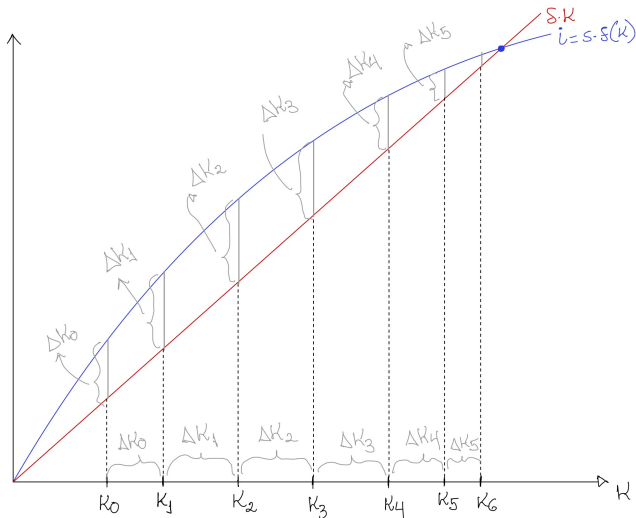
The capital stock at time $t = 2$ is:

$$k_2 = k_1 + \Delta k_1 = k_1 + sf(k_1) - \delta k_1$$

And so on...

Given k_0 , we can determine the capital stock at any point in time, by applying the equation above successively.

Let's see how the capital stock evolves over time:



It seems that the capital stock is slowly converging to a point. What is that point?
That point is the **steady state** of the economy.

Steady State

Steady state:

- It is the point where the **capital stock does not change** over time.
- The investment in the economy is just enough to replace the capital that wears out.
- We will refer to it as k^* .
- At the steady state, $\Delta k^* = 0$.

This gives us an expression to find the steady state:

$$\Delta k^* = sf(k^*) - \delta k^* = 0 \implies$$

$$\boxed{sf(k^*) = \delta k^*}$$

What determines the steady state?

- The **savings rate** s .
- The **depreciation rate** δ .
- The **production function** $f(k)$.

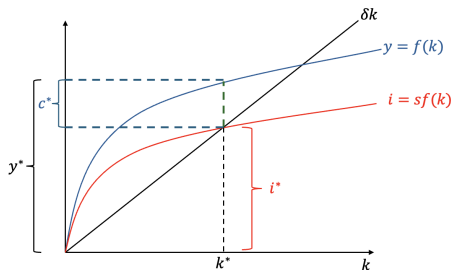
Steady State

We will refer to the variables at the steady state with a star!

At the steady state, the capital stock doesn't change, then:

- **Output** per worker is **constant**: $y^* = f(k^*)$
- **Investment** per worker is **constant**: $i^* = s \cdot y^*$
- **Consumption** per worker is **constant**: $c^* = (1 - s)y^*$

You can find the steady state graphically:



Steady State

Let's see how changes in the savings rate and the depreciation rate affect the steady state: [Steady State Solow Model App](#)

A **decrease** in the **depreciation rate**:

- **Increases** the steady state **capital** stock.
- **Increases** the steady state **output** per worker.
- **Increases** the steady state **consumption** per worker.

What about an **increase** in the **savings rate**?

- **Increases** the steady state **capital** stock.
- **Increases** the steady state **output** per worker.
- The effect on **consumption** per worker is **ambiguous**. Why?

Steady State

The effect on consumption is **ambiguous** because: $c^* = (1 - s) \cdot y^*$

- An **increase** in the savings rate increases output per worker.
- But it also **decreases** the **share** of output that people **consume** $1 - s$.

Conclusion: higher savings rates lead to higher output per worker, but not necessarily to higher consumption per worker!

Why do we care so much about the steady state? Two reasons:

- An economy at the steady state **will stay there forever**, unless something changes.
- An economy not at the steady state **will move towards it!**
 - **No matter the level** of **capital** with which the economy **begins**, it ends up with the steady-state level of capital!
 - That's why the steady state represents the **long-run equilibrium** of the **economy**.

Saving rate and growth in the Solow Model

This is really important:

An economy not at the steady state will move towards it!

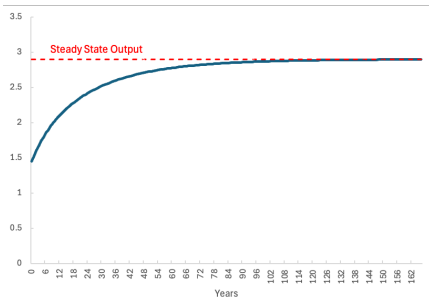
Let's see how the economy moves towards the steady state.

The following plot shows the evolution of output per capita in the Solow model, with the following parameters:

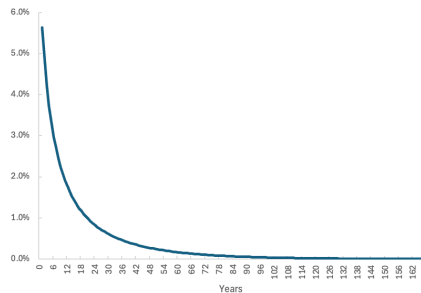
- $s = 0.6$
- $\delta = 0.05$
- $f(k) = k^{0.3}$
- $k_0 = 3.48$

Savings rate and growth in the Solow Model

Output per capita



Growth rate of output per capita



The economy is growing, but the growth rate is decreasing!

The **growth rate approaches zero** as the economy approaches the steady state.

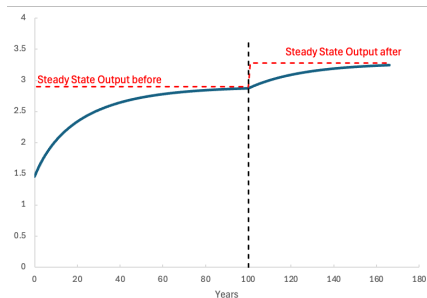
Can we have sustained growth in this Solow model?

Savings rate and growth in the Solow Model

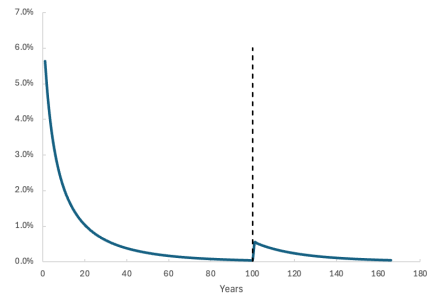
We could try increasing the savings rate.

Let's increase the saving rate to 0.8 at year 100 and see what happens.

Output per capita



Growth rate of output per capita



The growth rate **jumps up** but then **decreases** and **approaches zero** as the economy approaches the new steady state.

Savings rate and growth in the Solow Model

The increase in the growth rate is temporary!

We could try increasing the savings rate again and again.

But we can't do this forever. Why?

We can't have a savings rate greater than 1!

- People can't save more than their income!

Conclusion: We can't have sustained growth in the Solow model.

We will have to introduce population growth and technological progress to have sustained growth. Next Lecture!

Golden Rule

We saw that the effect of the savings rate on consumption per worker is ambiguous.

Let's think about a policymaker that wants to maximize economic well-being by choosing the savings rate.

What people really care about is **consumption**, not output.

So the policymaker should choose the **savings rate** that **maximizes consumption** per worker.

This is the **Golden Rule** savings rate.

Let's play a bit with the savings rate here: [Golden Rule Solow App](#)

Golden Rule

We will now see how to find the Golden Rule savings rate.

We will follow a 2-step process:

1. Find the capital per worker that maximizes consumption per worker, at the steady state. This is called the Golden rule level of capital.
2. Find the savings rate that leads to that capital per worker.

Let's start with the first step:

Consumption at the steady state is given by:

$$c^* = \underbrace{f(k^*)}_{y^*} - i^*$$

Now, at the steady state investment is equal to depreciation, then:

$$c^* = f(k^*) - \delta k^*$$

We want k^* that maximizes c^* . We will call it k_g .

Golden Rule

How do we find k_g ? **First order condition!**

Take the derivative of c^* with respect to k^* and set it equal to zero:

$$f'(k_g) - \delta = 0 \implies$$

$$\boxed{MPK = f'(k_g) = \delta}$$

The **capital** per worker that **maximizes consumption** per worker is the one that makes the **MPK equal** to the **depreciation rate**.

Second step: find the savings rate that leads to k_g .

At the steady state, investment is equal to depreciation. Then:

$$sf(k_g) = \delta k_g \implies$$

$$\boxed{s_g = \frac{\delta k_g}{f(k_g)}}$$

Golden Rule

Example: The production function is given by a Cobb-Douglas $F(K, L) = A \cdot K^\alpha L^{1-\alpha}$. The depreciation rate is δ . Find the Golden Rule savings rate. **Answer:** $s_g = \alpha$

⚠ The economy **does not** automatically **gravitate toward** the **Golden Rule savings rate**.

- If we want a particular k^* , such as the golden one, we need a saving rate that leads to that k^* .

Now, is it easy for this policymaker to implement the Golden Rule savings rate?

Will people support this policy?

As it is always the case in economics, the answer is: **it depends!**

The transition to the Golden Rule steady state

Suppose the economy has reached a steady state other than the Golden Rule.

A policymaker sets a new savings rate that leads to the Golden Rule steady state.

What happens to consumption, investment and capital when the economy transition from the old steady state to the new one?

We will have to consider two cases:

- The **current savings rate is higher** than the Golden Rule savings rate.
- The **current savings rate is lower** than the Golden Rule savings rate.

The transition to the Golden Rule steady state

Let's start with the **first case**: the current savings rate is higher than the Golden Rule savings rate.

We will call t_0 the time when the policymaker implements the golden rule savings rate.

The economy is in the old steady state and, at time t_0 , the savings rate is **reduced** to the Golden Rule savings rate.

Let's analyze what happens to all variables following this change.

The transition to the Golden Rule steady state

At t_0 , there's an immediate decrease in investment and, consequently, an immediate increase in consumption.

Now, investment is lower than depreciation $\implies$ We are no longer in the steady state.

Capital slowly decreases, leading to reductions in output, consumption, and investment.

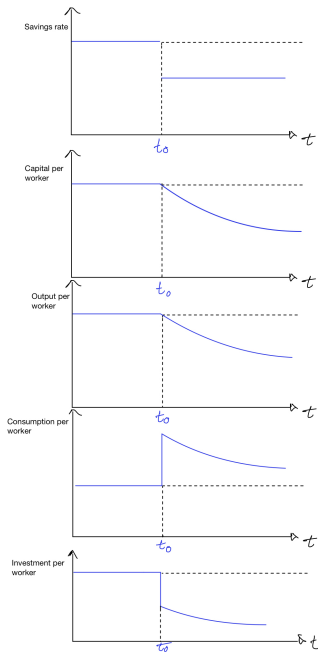
These variables decrease until a new steady state is reached.

Since we are assuming the new steady state is the golden rule, consumption in the new steady state must be higher than it was in the old steady state.

Conclusion:

- Consumption is higher at any point in time after the change in the savings rate.
- Easy to convince people to support this policy.

Transition to the Golden Rule steady state



Transition to the Golden Rule steady state

Now, let's analyze the **second case**: the current savings rate is lower than the Golden Rule savings rate.

Again, we will call t_0 the time when the policymaker implements the golden rule savings rate.

The economy is in the old steady state and, at time t_0 , the savings rate is **increased** to the Golden Rule savings rate.

Let's analyze what happens to all variables following this change.

Transition to the Golden Rule steady state

At t_0 , there's an immediate increase in investment and, consequently, an immediate decrease in consumption. Not good =(

Now, investment is higher than depreciation $\implies$ We are no longer in the steady state.

Capital slowly increases, leading to increases in output, consumption, and investment.

These variables increase until a new steady state is reached.

Again, the new steady state is the golden rule, then consumption in the new steady state must be higher than it was in the old steady state!

Conclusion:

- Consumption is higher in the very long-run, but lower immediately after the change in the savings rate.
 - Trade-off between reduced consumption now and higher consumption later.
- It might not be easy to convince people to support this policy.
 - Policymaker might have to sacrifice consumption of current generations for future generations.

Transition to the Golden Rule steady state

